

Total Questions: 35

LEVEL – II

SET - B

Time: 1.5 hr.
Total Marks: 70

SECTION - 1	- This section contains FIFTEEN (15) questions. Each question has FOUR options. ONLY ONE of these four options is the correct answer. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers.
SECTION - 2	- This section contains EIGHT (08) questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s). - For each correct answer you will get TWO marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
SECTION - 3	- This section contains NINE (09) questions of PASSAGE TYPE OR COMPREHENSION TYPE. - Under each paragraph there will be three questions that has FOUR options. ONLY ONE of these four options is the correct answer. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for
SECTION - 4	- This section contains THREE (03) questions. These questions are MATCHING TYPE BASED QUESTIONS. - For each correct answer you will get TWO marks. There are NO NEGATIVE marks for wrong answers

Section - 1

- The total number of seats in rows of an auditorium increases in A.P. If the first row has 20 seats and the last row has 50 seats, and there are 16 rows, find the total number of seats.
(A) 560 (B) 540
(C) 460 (D) 563
- If the roots of a quadratic equation are α and β such that $\alpha + \beta = 4$ and $\alpha\beta = 3$, then the equation is:
(A) $x^2 - 4x + 3 = 0$ (B) $x^2 + 4x + 3 = 0$
(C) $x^2 - 3x + 4 = 0$ (D) $x^2 + 3x + 4 = 0$
- A point P divides the line joining A(2, 4) and B(8, 10) internally in the ratio 1 : 2. What are the coordinates of P?
(A) (4, 6) (B) (6, 8)
(C) (5, 7) (D) (3, 5)
- When a polynomial is divided by $(x - 2)(x - 3)$, the remainder is $ax + b$. If $f(2) = 5$ and $f(3) = 7$, find the remainder.
(A) $x + 3$ (B) $2x + 1$
(C) $x + 1$ (D) $3x - 1$

5. Two equations represent the cost of apples and oranges:

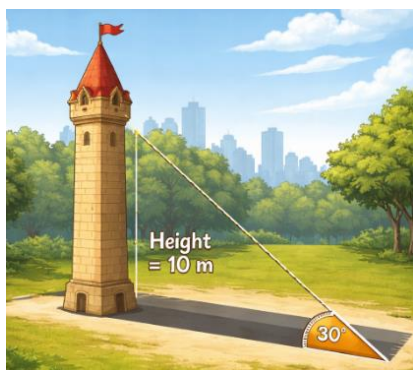
2 apples + 3 oranges cost ₹26

4 apples + 1 orange cost ₹18

Find the cost of one apple.

- (A) ₹4.5 (B) ₹2.8
(C) ₹3 (D) ₹6

6. A tower casts a shadow such that the angle of elevation of the top from the end of the shadow is 30° . If the height of the tower is 10 m, what is the length of the shadow?



- (A) $10\sqrt{3}$ (B) $\frac{10}{\sqrt{3}}$
(C) $5\sqrt{3}$ (D) 20

7. If the distance of a chord from the centre increases, what happens to its length?

- (A) Increases
(B) Decreases
(C) Remains same
(D) Becomes diameter

8. A bacteria culture doubles every hour. If initially there are 5 bacteria, how many will be there after 4 hours?

- (A) 20 (B) 40
(C) 80 (D) 160

9. A cone and cylinder have same radius and height. If cylinder volume is 300 cm^3 , what is the cone volume?

- (A) 100 cm^3 (B) 150 cm^3
(C) 200 cm^3 (D) 300 cm^3

10. Two triangles have equal altitudes. If the ratio of their bases is 3: 5, and the sum of their areas is 128 cm^2 , what is the area of the larger triangle?

- (A) 80 cm^2 (B) 64 cm^2
(C) 96 cm^2 (D) 100 cm^2

11. The mean of 15 numbers is 24. Five numbers are removed, each equal to 30, and the remaining numbers are increased by 3. What is the new mean?

- (A) 21 (B) 24
(C) 27 (D) 30

12. A school distributes 437 chocolates equally among some students such that each student gets 12 chocolates and some chocolates are left. How many chocolates are left?

- (A) 1 (B) 5
(C) 7 (D) 11

13. Two fair coins are tossed three times. What is the probability that exactly two times you get one head and one tail?

- (A) $\frac{3}{8}$ (B) $\frac{3}{4}$
(C) $\frac{9}{16}$ (D) $\frac{1}{2}$

14. The angle of elevation of the top of a building from a point on the ground is 60° . If the point is 20 m away from the building, find the height of the building.



- (A) $20\sqrt{3}$ m (B) $\frac{20}{\sqrt{3}}$ m
(C) $10\sqrt{3}$ m (D) 40 m
15. If $(x-2)$ and $(x-3)$ are factors of $f(x) = x^3 + ax^2 + bx + 6$, find $a + b$.
- (A) -5 (B) -3
(C) -7 (D) -9

Section - 2

16. The number of seats in rows of an auditorium increases by 3 in each row. The first row has 15 seats. Which of the following are correct?
- (A) Number of seats in 10th row = 42
(B) Total seats in first 10 rows = 285
(C) It forms an AP with $d = 3$
(D) Number of seats in n th row = $15 + (n - 1) \times 3$
17. A robot moves between points A (2, 3) and B(8, 3) on a horizontal path. Which of the following statements are correct?

- (A) Distance AB = 6 units
(B) Distance AB equals the length of the segment joining (2, 0) and (8, 0)
(C) Distance AB = $\sqrt{36}$
(D) Distance AB = 5 units

18. A fair die is thrown twice. Define the events:

E: "The sum of the numbers is a multiple of 3"

F: "At least one number is less than 3"

Which of the following are correct?

- (A) $P(E) = \frac{12}{36}$ (B) $P(F) = \frac{20}{36}$
(C) $P(E \cap F) = \frac{8}{36}$
(D) E and F are independent

19. A taxi service charges a fixed booking fee and a constant rate per km.

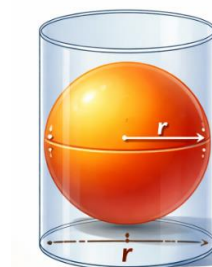
Two customers report:

- 5 km $\rightarrow$ ₹80
- 9 km $\rightarrow$ ₹128

Which of the following statements are correct?

- (A) The situation can be modeled using a pair of linear equations
(B) Cost per km is ₹12
(C) Fixed charge is ₹20
(D) Total fare for 15 km is ₹200

20. A spherical ball of radius r is placed inside a cylinder such that the cylinder has the same radius and height equal to the diameter of the sphere.



Which of the following statements are correct?

- (A) Height of cylinder = $2r$
- (B) Curved surface area of cylinder equals surface area of sphere
- (C) Volume of sphere equals volume of cylinder
- (D) Radius of cylinder equals radius of sphere

21. The difference between two positive numbers is 3. The sum of their reciprocals is $\frac{5}{18}$. Which of the following are correct?

- (A) If the larger number is x , then $\frac{1}{x} + \frac{1}{x-3} = \frac{5}{18}$
- (B) The equation simplifies to $5x^2 - 15x - 54 = 0$
- (C) The numbers are 9 and 6
- (D) The product of the numbers is 54

22. A person saves ₹100 more each month than previous month. First month saving ₹500.

Which are correct?

- (A) Savings form A.P
- (B) Common difference = 100
- (C) 5th month saving = 900
- (D) 6th month saving = 1000

23. Consider the pair of linear equations:

$$2x + y = 7, \quad 4x + 2y = 20$$

Which of the following statements are correct?

- (A) The system has no solution
- (B) The system is inconsistent
- (C) The lines represented by the equations are parallel
- (D) The system has infinitely many solutions

Section - 3

PASSAGE 1:

Building Observation

A person stands on top of a building of height **40 m**. He observes two points **A** and **B** on the ground in the same straight line and same side of the building.

The angle of depression of point **A** is 60° and that of point **B** is 30° .

24. What is the horizontal distance of point **A** from the base of the building?

- (A) $\frac{40}{\sqrt{3}}$ m
- (B) $40\sqrt{3}$ m
- (C) 20 m
- (D) 80 m

25. If the height of the building is increased to **80 m**, what will be the new distance of point **A** from the base?

- (A) $\frac{80}{\sqrt{3}}$ m
- (B) $80\sqrt{3}$ m
- (C) 40 m
- (D) 80 m

26. What is the horizontal distance of point **B** from the base of the building?

- (A) $\frac{40}{\sqrt{3}}$ m
- (B) $40\sqrt{3}$ m
- (C) 80 m
- (D) 20 m

PASSAGE-2

A rectangular field has an area of 96 m^2 .

The length of the field is 4 m more than its breadth. The dimensions of the field satisfy a quadratic equation.

27. If breadth is x , which equation is formed?

- (A) $x^2 + 4x - 96 = 0$
- (B) $x^2 - 4x - 96 = 0$
- (C) $x^2 + 4x + 96 = 0$
- (D) $x^2 - 96 = 0$

28. Find the breadth of the field.

- (A) 8 m (B) 6 m
(C) 12 m (D) 4 m

29. Find the length of the field.

- (A) 10 m (B) 12 m
(C) 14 m (D) 8 m

PASSAGE-3

A shopkeeper prepares an ice-cream cone using a conical mould.

The cone has a radius of 3 cm and a vertical height of 4 cm.

He wants to calculate the material needed to cover the outer surface and also the quantity of ice cream it can hold.



Use this information to answer the following questions.

30. To determine the material required to cover the cone, the shopkeeper first needs the slant height of the cone. What is the slant height?

- (A) 4 (B) 5
(C) 6 (D) 7

31. The shopkeeper wants to cover only the curved surface of the cone with chocolate coating. What is the curved surface area of the cone?

- (A) 12π (B) 15π
(C) 20π (D) 24π

32. To fill the cone with ice cream, the shopkeeper needs to know its capacity. What is the volume of the cone?

- (A) 12π (B) 24π
(C) 36π (D) 48π

Section - 4

33. A GPS system divides routes and calculates distances.

Column I (Situation)	Column II (Result)
(a) Point dividing A(2,2), B(10,10) in ratio 1:3	(i) (4,4)
(b) External division of A(2,2), B(6,6) in ratio 1:2	(ii) (-2,-2)
(c) Midpoint of A(4,-2), B(8,6)	(iii) (6,2)
(d) Distance between A(-1,-2) and B(5,4)	(iv) $\sqrt{72}$

- (A) a-i, b-ii, c-iii, d-iv
(B) a-ii, b-i, c-iii, d-iv
(C) a-i, b-ii, c-iv, d-iii
(D) a-iii, b-ii, c-i, d-iv

34. A student draws a card from a well-shuffled deck of 52 playing cards and records probabilities of different events

Column I (Situation)	Column II (Result)
(a) Drawing a red card	(i) $\frac{4}{52}$
(b) Drawing a king	(ii) $\frac{13}{52}$
(c) Drawing a face card	(iii) $\frac{3}{13}$
(d) Drawing a spade	(iv) $\frac{1}{2}$

- (A) a-i, b-ii, c-iii, d-iv
 (B) a-iv, b-i, c-iii, d-ii
 (C) a-i, b-ii, c-iv, d-iii
 (D) a-iii, b-ii, c-i, d-iv

35. A teacher gives quadratic equations arising from different applications and asks students to classify them based on discriminant.

Column I (Equations)	Column II (Discriminant Type)
(a) $x^2 - 6x + 9 = 0$	(i) $D = 0$
(b) $x^2 - 4x + 3 = 0$	(ii) $D > 0$
(c) $x^2 + 2x + 5 = 0$	(iii) $D < 0$
(d) $2x^2 - 7x + 3 = 0$	(iv) $D = 25$

- (A) a-i, b-ii, c-iii, d-iv
 (B) a-ii, b-i, c-iii, d-iv
 (C) a-i, b-ii, c-iv, d-iii
 (D) a-iii, b-ii, c-i, d-iv